

CA2355: Unsafe DataSet or DataTable in deserialized object graph (code analysis)

CA2355: Unsafe DataSet or DataTable in deserialized object graph (code analysis) - dotpaul, learn.microsoft.com.

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Content

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CA2355: Unsafe DataSet or DataTable in deserialized object graph

Summarize this article for me

Property Value	Rule ID CA2355	Title Unsafe DataSet or DataTable in deserialized object graph	Category Security	Fix is breaking or non-breaking Non-breaking	Enabled by default in .NET 10 No	Applicable languages C# and Visual Basic
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Cause

Deserializing when the casted or specified type's object graph can include a [DataSet](#) or [DataTable](#).

This rule uses a different approach to a similar rule, [CA2353: Unsafe DataSet or DataTable in serializable type](#).

The casted or specified type is evaluated when:

- Initializing a [DataContractSerializer](#) object
- Initializing a [DataContractJsonSerializer](#) object
- Initializing an [XmlSerializer](#) object

- Invoking [JavaScriptSerializer.Deserialize](#)
- Invoking [JavaScriptSerializer.DeserializeObject](#)
- Invoking [XmlSerializer.FromTypes](#)
- Invoking `Newtonsoft.Json.NET.JsonSerializer.Deserialize`
- Invoking `Newtonsoft.Json.NET.JsonConvert.DeserializeObject`

Rule description

When deserializing untrusted input with [BinaryFormatter](#) and the deserialized object graph contains a [DataSet](#) or [DataTable](#), an attacker can craft a malicious payload to perform a denial of service attack. There may be unknown remote code execution vulnerabilities.

For more information, see [DataSet and DataTable security guidance](#).

How to fix violations

- If possible, use [Entity Framework](#) rather than [DataSet](#) and [DataTable](#).
- Make the serialized data tamper-proof. After serialization, cryptographically sign the serialized data. Before deserialization, validate the cryptographic signature. Protect the cryptographic key from being disclosed and design for key rotations.

When to suppress warnings

It's safe to suppress a warning from this rule if:

- You know the input is trusted. Consider that your application's trust boundary and data flows may change over time.
- You've taken one of the precautions in [\[How to fix violations\]\(\)](#).

Suppress a warning

If you just want to suppress a single violation, add preprocessor directives to your source file to disable and then re-enable the rule.

```
#pragma warning disable CA2355
// The code that's violating the rule is on this line.
#pragma warning restore CA2355
```

To disable the rule for a file, folder, or project, set its severity to `none` in the [configuration file](#).

```
[*.{cs,vb}]
dotnet_diagnostic.CA2355.severity = none
```

For more information, see [How to suppress code analysis warnings](#).

Pseudo-code examples

Violation

```
using System.Data;
using System.IO;
using System.Runtime.Serialization;

[Serializable]
public class MyClass
{
    public MyOtherClass OtherClass { get; set; }
}

[Serializable]
public class MyOtherClass
{
    private DataSet myDataSet;
}

public class ExampleClass
{
    public MyClass Deserialize(Stream stream)
    {
        BinaryFormatter bf = new BinaryFormatter();
        return (MyClass) bf.Deserialize(stream);
    }
}
```

CA2350: Ensure `DataTable.ReadXml()`'s input is trusted

CA2351: Ensure `DataSet.ReadXml()`'s input is trusted

CA2352: Unsafe `DataSet` or `DataTable` in serializable type can be vulnerable to remote code execution attacks

CA2353: Unsafe `DataSet` or `DataTable` in serializable type

CA2354: Unsafe `DataSet` or `DataTable` in deserialized object graph can be vulnerable to remote code execution attack

CA2356: Unsafe `DataSet` or `DataTable` in web deserialized object graph

CA2361: Ensure autogenerated class containing `DataSet.ReadXml()` is not used with untrusted data

CA2362: Unsafe DataSet or DataTable in autogenerated serializable type can be vulnerable to remote code execution attacks

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